

Problem T. Implement MinHeap

OS Linux

Perform Insert, delMin and getMin operations on a Min Heap. Implement Min Heaps and avoid using inbuilt library.

Input Format

First line of input contains T - number of operations. Its followed by T lines, each line contains either "insert x" or "delMin" or "getMin".

Constraints

$$1 \leq T \leq 10^6$$

$$-10^9 \leq x \leq 10^9$$

Output Format

For each "getMin" operation, print the minimum element, separated by newline. If the heap is empty, print "Empty".

Input	Output
10 insert 5 getMin delMin getMin insert -15 insert 10 insert -30 getMin delMin getMin	5 Empty -30 -15

Explanation 0

Self Explanatory